

Software Foundation. Octave is written in C++ and was first used in chemical reactor design. Later, it got redeveloped for academic, research and commercial applications. The software is built using standard numerical libraries like LAPACK (linear algebra package) and BLAS (basic vector and matrix operations). Octave resembles MATLAB closely, in terms of its feature set and the syntax that it uses. It primarily features linear and matrix algebra, statistics, geometry, sets, object-oriented programming, polynomial manipulations, as well as signal, image and audio processing. As a programming language, it is interpreted and follows a structured programming style similar to C with access to standard library functions and certain UNIX system calls. Its interpreter has an open graphics library (OpenGL) based graphics engine to create plots, graphs and charts. Octave also gets well with popular graphing utilities like Gnuplot and Plplot. Just like Scilab, Octave is considered as another great alternative to MATLAB.

It is available for download at <https://www.gnu.org/software/octave/download.html>.

FreeMat

Another open source option for Windows users is FreeMat, which comes with an interactive shell and is famous for its 3D visualisation and plotting capabilities in a way similar to Matplotlib and R. The software extensively uses the OpenGL library that renders 3D functionalities. FreeMat features an interface to programming languages like C, C++ and Fortran. Users can download it from <http://freemat.sourceforge.net/download.html>.

SageMath

SageMath, shortened to Sage, is mathematical software

written in Python for algebra, geometry, number theory, cryptography, and various other related mathematical fields. The official Sage documentation states that the software is not natively available on Windows but can be run with the help of a virtual machine or the Cygwin Linux API layer. However, as Sage is popular in the field of computations, I felt it worthwhile to cover it in this article. Its beauty is that besides its core functionalities, it builds on top of nearly a hundred other established open source mathematical packages like SymPy, SciPy, Maxima, NumPy, R, GAP, etc, thereby making it a rich collection of functionalities and a superset of various packages. Users can interact with Sage through an interactive shell or Jupyter notebook, which is a Web UI that Python users are familiar with. The software also comes with advanced mathematical functionalities like Dirichlet characters, group theory, algebraic geometry, differential equations, multivariate polynomial gcd and discrete algebra. Interested readers can refer to the official Sage manual to explore all that it has to offer. **END**

References

- [1] <http://www.scilab.org/>
- [2] <https://www.gnu.org/software/octave/about.html>
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